

Shiva Upadhye

Curriculum Vitae

CONTACT INFORMATION

Shiva.Upadhye@tufts.edu
Department of Psychology
Tufts University
490 Boston Ave, Medford, MA 02155

Appointments

Postdoctoral Scholar August 2026 - Present
Department of Psychology, Tufts University
NeuroCognition of Language Lab (PI: Gina Kuperberg)

Education

University of California, Irvine June 2026
Ph.D. in Language Science
Dissertation: *Information and Performance Constraints in Online Language Production*
Committee: Richard Futrell (chair), Gregory Hickok, Gregory Scontras, Xin Xie, and Victor Ferreira

University of California, San Diego June 2020
Bachelor of Science in Cognitive Science
Concentration: Machine Learning and Neural Computation
Minor: Mathematics

Publications

Baidildinova M., **Upadhye S.**, Wagner A., Mayer M., & Futrell, R. (2025). Quantifying Mutual Intelligibility using Language Models: Evidence from Turkic Languages. *Proceedings of the Society of Computation in Linguistics*.

Upadhye S. & Futrell, R. (2025). Back to the Future: The Role of Past and Future Context Predictability in Incremental Language Production. *(Revise and Resubmit)*.

Upadhye S. Li Y., Li J. Attali, N. & Scontras G. (2025). GPT-2's Production and Comprehension of Scope Ambiguity (2025). *PragLM @ Conference on Language Modeling*.

Upadhye S. & Futrell, R. (2025). Examining Future Context Predictability Effects in Word-form Variation and Word Choice. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*.

Upadhye S., Li J., & Futrell, R. (2025). SPACER: A Parallel Dataset of Speech Production And Comprehension of Error Repairs. *The 14th edition of the Workshop on Cognitive Modeling and Computational Linguistics*.

Upadhye S. & Futrell, R. (2023). Typing time PWI: A scalable paradigm for studying lexical production. *Proceedings of the 45th Annual Meeting of the Cognitive Science Society*.

Upadhye, S. Li, J., & Futrell, R. (2023) Bridging production and comprehension: A computational model of error generation and correction. *Proceedings of the Society*

for Computation in Linguistics.

Upadhye S. & Futrell, R. (2022). Information-theoretic analyses of disfluencies in Speech. *NeurIPS 2022 Workshop on Information-Theoretic Principles in Cognition Systems.*

Upadhye S., Bergen L., & Kehler, A (2020). Predicting Reference: What do Language Models learn about Discourse Models? *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing.*

**Conference
Presentations**

Upadhye S. & Futrell, R. (2026). Repairing, Fast and Slow: Lexical and Contextual Pressures on the Timecourse of Error Processing in Naturalistic Production. *The 39th Annual Conference on Human Sentence Processing.* Cambridge, MA, USA. **(Poster)**.

Baidildinova, M., **Upadhye S.**, Wagner, A., Mayer, C., & Futrell, R. (2026). Quantifying Mutual Intelligibility using Language Models: Evidence from Turkic Languages. *The 39th Annual Conference on Human Sentence Processing.* Cambridge, MA, USA. **(Poster)**.

Upadhye S., Li Y., Li J. Attali, N. & Scontras G. (2025). GPT-2's Production and Comprehension of Scope Ambiguity (2025). *PragLM @ Conference on Language Modeling.* Montreal, Quebec, Canada. **(Poster)**

Upadhye S., Yeaton, JD., Yi, E., & Hickok, G. (2025). Speakers and typers adapt their planning strategies to meet sentence production demands. *66th Annual Meeting of the Psychonomics Society.* Denver, CO, USA. **(Poster)**

Upadhye S. & Futrell, R. (2025). Examining Future Context Predictability Effects in Word-form Variation and Word Choice (2025). *47th Annual Meeting of the Cognitive Science Society.* San Francisco, CA, USA. **(Talk)**

Upadhye S., Yeaton, JD., Yi, E., & Hickok, G. (2025). Typing Fluency reveals Psycholinguistic Dynamics of Sentence Production. *The 38th Annual Conference on Human Sentence Processing.* College Park, MD, USA. **(Poster)**

Li J., **Upadhye S.**, Gandhasri, R., & Futrell, R. (2025). SPACER: A Parallel Dataset of Speech Production And Comprehension of Error Repairs. *The 38th Annual Conference on Human Sentence Processing.* College Park, MD, USA. **(Poster)**

Li Y., Li J., **Upadhye S.**, & Scontras, G. (2025). Language models prefer ambiguous utterances following informative contexts. *The 38th Annual Conference on Human Sentence Processing.* College Park, MD, USA. **(Poster)**

Yeaton, JD, **Upadhye, S.**, Futrell, R., & Hickok, G. (2024). The (Non-)Sentence Repetition Task: A novel assessment of expressive and receptive syntactic competence. *16th Annual Meeting of the Society for the Neurobiology of Language.* Brisbane, Australia. **(Poster)**

Upadhye S., Attali N. & Scontras, G. (2023). The role of world knowledge for interpreting aspect ambiguity. *Tenth Biennial Meeting of Experimental Pragmatics.* Paris, France. **(Poster)**

Li, J., **Upadhye, S.** & Futrell, R. (2023) Bridging production and comprehension: A computational model of error generation and correction. *The 36th Annual Conference*

on Human Sentence Processing. Pittsburgh, PA, USA. (**Poster**)

Upadhye S. & Futrell R. (2022). Information-theoretic analysis of disfluencies in speech. *NeurIPS 2022 Workshop on Information Theoretic Principles in Cognitive Systems*. New Orleans, LA, USA. (**Poster**)

**Other
Contributions**

Moslehi B., Price W., Moslehi B., Off J., Sourichanh K., Cheng Y., **Upadhye S.**, & Soutoudeh V. (2025). *Dual cavity interferometer sensor system*. U.S. Patent No. US20250224295A1. US Patent & Trademark Office.

**Teaching &
Mentorship**

UC Irvine
Instructor of Record
Introduction to Linguistics

Summer 2025

Pedagogical Training

Course Development Fellowship for Language Science (2024)

UCI Division of Teaching Excellence & Innovation - Graduate Summer Fellowship & Certificate for Hybrid Inclusive Pedagogy (2022)

Teaching Assistant

2022 - 2025

Introduction to Linguistics, Research Methods in Psychology,
Psychology of Language, American Public Policy, Acquisition of Language, Discovering Language;

Mentorship

2023 - Current

Rutvik Mohan Gandhasri, Mackenzie Leigh Maxwell (UROP Fellow - 2024), Frankie Boren, Austin Wagner (UROP Fellow - 2025)

UC San Diego

Instructional Assistant
Computational Linguistics

Spring 2020

**Other
Experience**

AI & Data Analytics Engineer

Oct 2020 - July 2021

Intelligent Fiber Optic Systems Corporation

- Worked in cross-functional teams with physicists and engineers to develop signal processing solutions for fiber optic sensing systems.
- Contributed to the technical development of research proposals investigating applications of fiber-optic sensing solutions for renewable energy.

Computational Neurosciences Intern

Oct 2019 - Mar 2020

National Center for Microscopy and Imaging Research

- Principal Investigator: Dr. Matthias Haberl
- Contributed to the continued development of CDeep3M — a deep learning-based pipeline for automated segmentation of 3-D neuronal images.

Research Assistant & Undergraduate Researcher

Mar 2019 - June 2020

Computational Linguistics Lab at UC San Diego

- Principal Investigator: Dr. Andrew Kehler
- Conducted independent research investigating the role of verb biases and discourse structure knowledge on language models' pronominalization biases.
- Conducted corpus analysis to investigate naturalistic patterns of *sluicing*—a form of VP ellipsis.

**Service,
Fellowships, &
Awards**

Program Chair and Organizer

Society for Computation in Linguistics (SCiL 2024)

Reviewer

Annual Meeting of Cognitive Science Society (Cogsci), Society for Computation in Linguistics (SCiL), California Meeting on Psycholinguistics (CAMP), SoCal NLP

Awards & Fellowships

Associate Graduate Dean's Fellowship, UC Irvine (2021 - Present; 3 quarters)

UCI Center for Theoretical Behavioral Sciences (CTBS) Travel Award (2025)

Simmons Institute Analytical Connectionism Summer School Fellowship (2024; declined)

Intel Academic GPU Donation Program (2023) - Intel Arc A770 GPU